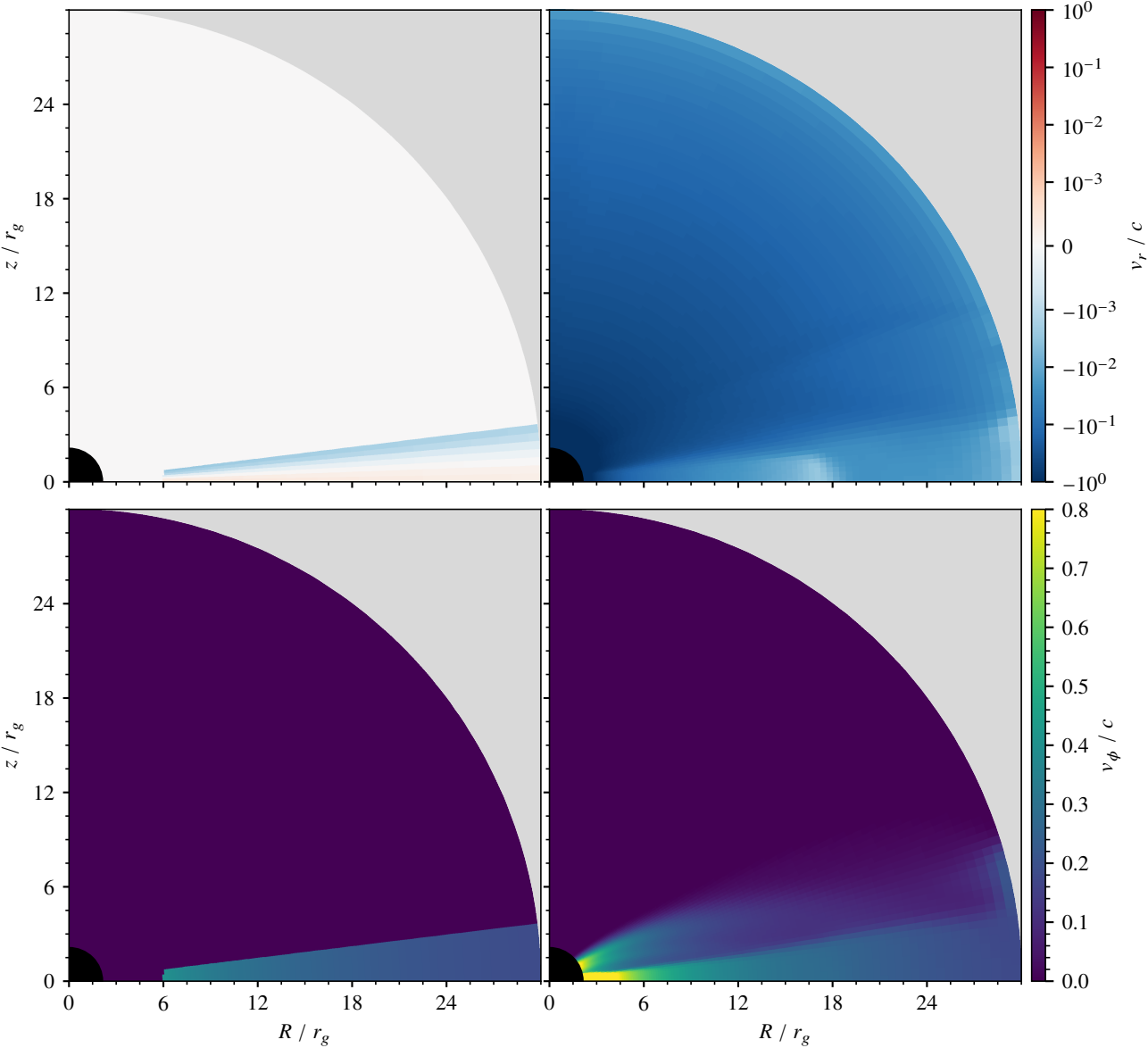

z / r_g
R / r_g
v_r / c
v_φ / c